Worksheet No. - 2

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1.Aim/Overview of the practical: Write a query to print the company_code, founder name, total number of lead managers, total number of senior managers, total number of managers, and total number of employees. Order your output by ascending company_code.

2.Objective: Create a table to store details of managers along with their associated senior managers, lead managers, and company information.

3.Procedure/Algorithm/Code:

First, I created a database:

```
create database corporate;
Query OK, 1 row affected (0.04 sec)
```

Then I used that database:

```
use corporate;
Database changed
```

Now, I created a company table with 2 attributes:

```
CREATE TABLE company (
    company_code VARCHAR(5),
    founder VARCHAR(20)
);
Query OK, 0 rows affected (0.08 sec)
```

I inserted 2 records into the company table:

```
insert into company values("c1", "Monika");
Query OK, 1 row affected (0.01 sec)
```

```
insert into company values("c2", "Samantha");
Query OK, 1 row affected (0.01 sec)
```

Next, I created a lead_manager table with 2 attributes:

```
CREATE TABLE lead_manager (
   lead_manager_code VARCHAR(7),
   company_code VARCHAR(7)
);
Query OK, 0 rows affected (0.02 sec)
```

I inserted 2 records into the lead_manager table:

```
mysql>  insert into lead_manager values("LM1", "c1");
Query OK, 1 row affected (0.01 sec)
mysql> insert into lead_manager values("LM2", "c2");
Query OK, 1 row affected (0.01 sec)
```

Then, I created a senior_manager table with 3 attributes:

```
CREATE TABLE senior_manager (
   senior_manager_code VARCHAR(10),
   lead_manager_code VARCHAR(10),
   company_code VARCHAR(10)
);
Query OK, 0 rows affected (0.03 sec)
```

I inserted 3 records into the senior_manager table:

```
mysql> insert into senior_manager values("SM1", "LM1", "c1");
Query OK, 1 row affected (0.01 sec)
mysql> insert into senior_manager values("SM2", "LM2", "c2");
Query OK, 1 row affected (0.00 sec)
mysql> insert into senior_manager values("SM3", "LM3", "c3");
Query OK, 1 row affected (0.01 sec)
```

After that, I created a manager table with 4 attributes:

```
CREATE TABLE manager (
   manager_code VARCHAR(10),
   senior_manager_code VARCHAR(10),
   lead_manager_code VARCHAR(10),
   company_code VARCHAR(10)
);
Query OK, 0 rows affected (0.02 sec)
```

I inserted 4 records into manager table:

```
mysql> insert into manager values("M1", "SM1", "LM1", "c1");
Query OK, 1 row affected (0.01 sec)
mysql>  insert into manager values("M2", "SM2", "LM2", "c2");
Query OK, 1 row affected (0.01 sec)
mysql>  insert into manager values("M3", "SM3", "LM2", "c2");
Query OK, 1 row affected (0.01 sec)
```

Finally, I created an employee table with 5 attributes:

```
mysql> CREATE TABLE employee (
        employee_code VARCHAR(10),
        manager_code VARCHAR(10),
        senior_manager_code VARCHAR(10),
        lead_manager_code VARCHAR(10),
        company_code VARCHAR(10)
       );
Query OK, 0 rows affected (0.03 sec)
```

I inserted 5 records into manager table:

```
mysql>  insert into employee values("E1", "M3", "SM1", "LM1", "c1");
Query OK, 1 row affected (0.01 sec)
mysql>  insert into employee values("E2", "M1", "SM1", "LM1", "c1");
Query OK, 1 row affected (0.01 sec)
mysql>  insert into employee values("E3", "M2", "SM3", "LM2", "c2");
Query OK, 1 row affected (0.01 sec)
mysql>  insert into employee values("E4", "M3", "SM3", "LM2", "c2");
Query OK, 1 row affected (0.01 sec)
```

```
SELECT
    c.company_code,
    c.founder,
    (SELECT COUNT(DISTINCT lead_manager_code)
    FROM lead_manager
   WHERE company_code = c.company_code) AS total_lead_managers,
   (SELECT COUNT(DISTINCT senior_manager_code)
    FROM senior_manager
   WHERE company_code = c.company_code) AS total_senior_managers,
   (SELECT COUNT(DISTINCT manager_code)
    FROM manager
   WHERE company_code = c.company_code) AS total_managers,
    (SELECT COUNT(DISTINCT employee_code)
    FROM employee
    WHERE company_code = c.company_code) AS total_employees
    FROM company c
    ORDER BY c.company_code ASC;
```

4.Output:

Company Table

company_code	founder
c1	Monika
c2	Samantha

Lead_manager Table

lead_manager_code	company_code
LM1	c1
LM2	c2

Senior_manager Table

senior_manager_code	lead_manager_code	company_code
SM1	LM1	c1
SM2	LM2	c2
SM3	LM3	c3

Manager table

manager_code	senior_manager_code	lead_manager_code	company_code
M1	SM1	LM1	c1
M2	SM3	LM2	c2
M3	SM3	LM2	c2

Employee Table

employee_code	manager_code	senior_manager_code	lead_manager_code	company_code
E1	M3	SM1	LM1	c1
E2	M1	SM1	LM1	c1
E3	M2	SM3	LM2	c2
E4	M3	SM3	LM2	c2

Final Output

company_code	founder	total_lead_managers	total_senior_managers	total_managers	total_employees
c1	Monika	1	1	1	2
c2	Samantha	1	1	2	2

Learning outcomes (What I have learnt):

1. Understand how to work with database relationships by linking multiple tables using common columns.
2. Practice using subqueries to aggregate data from related tables.
3. Learn how to sort query results using ORDER BY for clearer, organized output.